AMRIT RAJ

UNDERGRADUATE STUDENT

CONTACT

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Siwan, Bihar, India

SKILLS

- Technical Skills
- Languages: Python
- Libraries: NumPy, Matplotlib, Scikit-Learn
- Mathematics: Probability, Statistics, Combinatorics, Calculus, Algebra
- Methodologies: Machine-Learning, Modeling, Regression, Analytics, Visualization
- Domains: Electrodynamics, Genetics, Optics, Mechanics

LANGUAGES

- English
- Hindi

PROFILE

First-year BS-MS student at IISER Mohali combining a strong foundation in mathematical physics, probability theory, and electrodynamics with scientific computing. Passionate about leveraging Python (NumPy, Matplotlib) and machine learning frameworks like Scikit-Learn to model complex physical systems and analyze large experimental datasets

EDUCATION

Grade 10-12

Asarfilal High School, Siwan, Bihar 2022-2024

- Science Stream | IISER Aptitude Test (IAT) : AIR 1093

BS-MS Dual Degree 2025--2030

Indian Institute of Science Education and Research, Mohali

- CPI: 7.95 | Recent SPI: 8.74

Relevant Coursework & Independent Study

Computer Science & Machine Learning

- Introduction to Python & Theory of Computation
- Independent Study: MIT OCW 6.001 SC & 6.002 SC, Hands-on Machine Learning (O'Reilly)

Mathematics & Statistics

- Probability & Statistics, Introduction to Linear Algebra
- Differential & Integral Calculus, Sequences & Series
- Elementary Number Theory

Physics

- Overview of Mathematical Methods
- Classical Mechanics, Electromagnetism, Waves & Optics

LABORATORY SKILLS & EXPERIMENTAL TECHNIQUES

Physics & Instrumentation

- Mechanics & Matter: Analyzed simple harmonic motion using Kater's, compound, and torsion pendulums; measured 'g' via free fall; estimated Young's modulus utilizing Searle's method.
- Electromagnetism & Circuits: Conducted random sampling of AC sources and analyzed electromagnetic induction; mapped magnetic fields across 125mm, 100mm, and 75mm solenoids and circular coils; utilized oscilloscopes to analyze induction damping.
- Waves, Optics & Interferometry: Explored fundamental modes via sonometer and Melde's setup; analyzed sound waves on a CRO; operated prism and constant deviation spectrometers, polarimeters, Fresnel's biprisms, and laser diffraction gratings; measured specific wavelengths utilizing a Michelson interferometer and Newton's rings setup.

Biological Sciences

- Microbiology & Microscopy: Executed aseptic techniques, media preparation, bacterial culturing, serial dilution, cell counting (OD600 & Plating), and biological staining. Performed bright-field microscopy and sample preparation.
- In Vivo Techniques: Cultured, handled, and observed model organisms including Drosophila (Fruit Fly) and C. elegans.

Chemical Analysis

- Analytical Techniques: Performed UV-Vis Spectrophotometry for chlorophyll estimation, Thin Layer Chromatography (TLC), Column Chromatography, and Complexometric and Iodometric Titrations.
- Organic Chemistry: Conducted qualitative organic analysis, including observing absorption and effervescence reactions with $NaHCO_3$.